Project Document: NeuroHire - AI Resume Recruiter

1. Project Overview

NeuroHire is an AI-powered resume screening and job matching platform that analyzes a candidate's resume against multiple job descriptions. It leverages NLP, skill extraction, FAISS vector search, and intelligent rule-based recommendations to enhance recruitment precision.

2. Why This Project is Needed

Traditional resume screening is time-consuming, prone to bias, and often misses great candidates due to keyword mismatches. NeuroHire solves this by automating skill extraction, evaluating career progression, identifying gaps, and generating smart suggestions with confidence scores.

3. Differentiators from Other Platforms

- Multi-level analysis: From skills and education to role progression and red flags
- FAISS-powered job description matching engine
- ATS-style scoring and pie chart visualizations
- Suggestions for upskilling, culture fit, and career planning
- No dependency on fixed template resumes
- Streamlit-based fast, interactive UI

4. Key Modules

- Resume Parser (PDF reader, text extractor)
- JD Parser (manual or PDF upload)
- Skill Depth & Education Analyzer
- FAISS Semantic Matching Engine
- Screening & Tagging System
- Final Recommendation Generator

5. Future Roadmap

- Integration with LinkedIn / GitHub profile scraping

- AI-powered interview question generator
- Recruiter-side dashboard with filters and analytics
- NLP-based candidate clustering by role/tech stack
- Integration with Applicant Tracking Systems (ATS)

6. Tech Stack

- Streamlit (Frontend & UI)
- PyMuPDF (PDF parsing)
- Sentence-Transformers + FAISS (semantic search)
- Seaborn/Matplotlib (visuals)
- Python (NLP, logic, rule engine)
- HTML/CSS (styling)

7. Conclusion

NeuroHire is designed to streamline modern hiring, enabling more objective, skill-based evaluation while saving time and improving candidate-job fit. It is scalable, adaptable, and designed to evolve into a comprehensive AI assistant for recruiters.